

Sewing Constraint of Bulk and Boundary CFT

Yu Liu

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Abstract

This is a note capturing the main idea and calculation of [1], which focus on how sewing manifold will constraint the sturcture of a CFT with boundary presented.

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1 Bulk Sewing Constraint

1.1 Bulk Content

1.1.1 Data Convention

Here we set up the convention a Bulk CFT data. The correlation function of a bulk CFT can be defined with the following data:

- **Lower Point Function** Convention:

$$\begin{aligned}\langle \phi_i \rangle &= 0, & \langle I \rangle &= 1, \\ \langle \phi_i(z, \bar{z}) \phi_j(0, 0) \rangle &= \frac{\delta_{ij}}{(z\bar{z})^{2\Delta_i}},\end{aligned}\tag{1}$$

We take this convention of bulk 2 point function is always normalized. Then we define the bulk OPE coefficient C_{ij}^k as:

- **Bulk Field OPE Coefficient Convention:**

$$\phi_i(z, \bar{z})\phi_j(0, 0) = \sum_k C_{ij}^k \phi_k z^{\Delta_k - \Delta_i - \Delta_j} \bar{z}^{\Delta_k - \Delta_i - \Delta_j} + \text{Descendants...} \quad (2)$$

Then with this convention we can talk about the constraints these data should satisfy due to the consistency of the CFT.

There are 2 Bulk Sewing Constraints for a CFT. The first one is the crossing symmetry of the 4 point function. The second one is the modular invariance of the torus partition function. Now let's talk about the first one.

1.2 Crossing Symmetry

1.2.1 4 point function Decomposition

The 4 point function of a CFT can be written in the following way.

- First, the global Ward Identity tells us that the 4 point function can be written as some function of the cross ratio:

$$\langle \phi_1(z_1)\phi_3(z_2)\phi_3(z_3)\phi_4(z_4) \rangle = \prod_{i < j} (z_i - z_j)^{r - \Delta_i - \Delta_j} (\bar{z}_i - \bar{z}_j)^{\bar{r} - \bar{\Delta}_i - \bar{\Delta}_j} Y(\eta, \bar{\eta}), \quad (3)$$

where

$$r = \frac{1}{3} \left(\sum_i \Delta_i \right) \quad \bar{r} = \frac{1}{3} \left(\sum_i \bar{\Delta}_i \right) \quad (4)$$

and

$$\eta = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)} \quad (5)$$

is the cross ratio.

- Second, the OPE and Local Ward Identity gives us the following expressions of $Y(\eta, \bar{\eta})$:

$$\begin{aligned} Y(\eta, \bar{\eta}) &= \sum_k C_{12}^k C_{34}^k \mathcal{F}(12, 34; k)(\eta) \bar{\mathcal{F}}(12, 34; k)(\bar{\eta}) \\ &= \sum_k C_{14}^k C_{23}^k \mathcal{F}(14, 23; k)(1 - \eta) \bar{\mathcal{F}}(14, 23; k)(1 - \bar{\eta}) \\ &= \sum_k C_{13}^k C_{24}^k \mathcal{F}(13, 24; k) \left(\frac{1}{\eta} \right) \bar{\mathcal{F}}(13, 24; k) \left(\frac{1}{\bar{\eta}} \right) \end{aligned} \quad (6)$$

notice that:

$$\eta = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)} \quad 1 - \eta = \frac{(z_1 - z_4)(z_2 - z_3)}{(z_1 - z_3)(z_2 - z_4)} \quad \frac{1}{\eta} = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_2)(z_3 - z_4)} \quad (7)$$

which is the cross ratio of the 4 points in different channels.

1.2.2 Conformal Blocks

The special function above we use: $\mathcal{F}(12, 34; k)(\eta)$ is called the conformal block. It is a special function determined by the conformal symmetry. Here let's list out some basic properties of the conformal block:

1. These functions are **Chiral** and only depends on Chiral data. Thus, they are multivalued.

2. 4 point Conformal Blocks on a plane are multivalued and have branch cuts. The cutting points are at $\eta = 0, 1, \infty$. The conformal blocks are analytic on the complex plane with these cuts.

Remark

Indeed there are some subtleties about the convention, though we write i, j, k, l in the convention, it in fact only depends on the chiral data of the double fold virasoro primary.

The most important fact of the conformal blocks is that they satisfy some linear relations called duality relations. For 4 point blocks on a plane, its famous for the duality relation called the **F-move**:

$$\mathcal{F}(ij, kl; p)(\eta) = \sum_q F_{pq} \begin{pmatrix} i & l \\ j & k \end{pmatrix} \mathcal{F}(il, jk; q)(1 - \eta) \quad (8)$$

here $F_{pq} \begin{pmatrix} i & j \\ k & l \end{pmatrix}$ is called the F-matrix, which only depends on the chiral data of the CFT. They satisfies some consistency conditions which can eventually be determined analytically.

1.2.3 Crossing Symmetry

The crossing symmetry is the following statement:

- The 4 point function should be single valued and analytic on the whole complex plane. This means that the 3 different conformal blocks decomposition of the 4 point function should be the same function.

This statement can be written in a mathematical way as:

$$\begin{aligned} & \sum_p C_{12}^p C_{34}^p \mathcal{F}(12, 34; p)(\eta) \overline{\mathcal{F}}(12, 34; p)(\bar{\eta}) \\ &= \sum_q C_{14}^q C_{23}^q \mathcal{F}(14, 23; q)(1 - \eta) \overline{\mathcal{F}}(14, 23; q)(1 - \bar{\eta}) \end{aligned} \quad (9)$$

if we plug in the F-move, we can get:

Theorem 1.1

Crossing Symmetry

$$\sum_p C_{12}^p C_{34}^p F_{pq} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \overline{F}_{pl} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = C_{14}^q C_{23}^q \delta_{ql} \quad (10)$$

notice that the consistency condition of conformal blocks makes the F-matrix satisfy the following normalization condition:

$$\sum_q F_{pq} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \overline{F}_{ql} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = \delta_{pl} \quad (11)$$

Remark

Notice that $\overline{F}$ is the complex conjugate of F . In some cases the F-matrix is real, then we can just write F instead of $\overline{F}$.

Thus the crossing symmetry can be written in a more compact way as:

$$C_{12}^p C_{34}^p F_{pr} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = C_{14}^r C_{23}^r F_{rp} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \quad (12)$$

1.3 Modular Invariance

1.3.1 Modular S Invariance

Modular invariant is that the Torus partition function should be invariant under modular transformation. Particularly, we care most about the modular S transformation. The torus partition function invariant under the modular S transformation means that the partition function should satisfy the following condition:

$$Z(\tau) = Z\left(-\frac{1}{\tau}\right) \quad (13)$$

Torus partition function can be decomposed into characters:

$$Z(\tau) = \sum_{i,j} \chi_p(\tau) \bar{\chi}_p(\bar{\tau}) \quad (14)$$

Thus modular S invariance means that the characters should satisfy the following condition:

Theorem 1.2

Modular S Invariance

$$\sum_p \chi_p(\tau) \bar{\chi}_p(\bar{\tau}) = \sum_p \chi_p\left(-\frac{1}{\tau}\right) \bar{\chi}_p\left(-\frac{1}{\bar{\tau}}\right) \quad (15)$$

1.3.2 Character S Transformation

Just as the 4 point blocks transform under the F move in different channels. The CFT characters have following behavior:

$$\chi_i(\tau) = \sum_j S_{ij} \chi_j\left(-\frac{1}{\tau}\right) \quad (16)$$

where S_{ij} is called the S-matrix, which only depends on the chiral data of the CFT. In fact this follows from the fact that the torus partition function can be modular invariant.

2 Boundary Sewing Constraint

Now we admit the CFT to have a boundary on the real axis. Now we to restrict the conformal symmetry to preserve the boundary, which lead to the **conformal boundary condition**;

$$T(z) = \bar{T}(\bar{z}) \quad z = \bar{z} \quad (17)$$

with this boundary condition, the theory here only have 1 fold of Virasoro symmetry.

2.1 Boundary Content

2.1.1 Theory Contents

With the boundary presented, we need more data to characterize the CFT.

1. Boundary Conditions a
2. Boundary Operators $\psi_i^{(ab)}$, here i is the label of 1 fold Virasoro primary operator, and a, b are the labels of the boundary conditions.

2.1.2 Data Convention

A full CFT with boundary on UHP can be characterized by the following data:

- **Lower Point Function** Convention:

$$\langle \psi_i^{(ab)} \rangle = 0, \quad \langle I \rangle_a = \alpha^a \quad (18)$$

Notice that here, with the boundary presented and with the bulk convention fixed, we no longer can normalized the boundary 0 point function freely.

- **Boundary 2 Point Function** Convention:

$$\langle \psi_i^{(ab)}(0) \psi_j^{(cd)}(x) \rangle = \delta_{ij} \delta_{ad} \delta_{bc} x^{-2\Delta_i} \alpha_i^{(ad)}, \quad x > 0 \quad (19)$$

Indeed, we can choose to normalize the boundary field itself to make the boundary 2 point function to be normalized. However, this won't be a natural choice.

- **Boundary OPE** Coefficient Convention:

Bibliography

- [1] D. C. Lewellen, “Sewing Constraints for Conformal Field Theories on Surfaces with Boundaries,” *Nuclear Physics B*, vol. 372, no. 3, pp. 654–682, Mar. 1992, doi: [10.1016/0550-3213\(92\)90370-Q](https://doi.org/10.1016/0550-3213(92)90370-Q).